



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

**Circuits and Electronics Laboratory - ENEE2103
PreLab Exp5: Filters.**

Prepared By:

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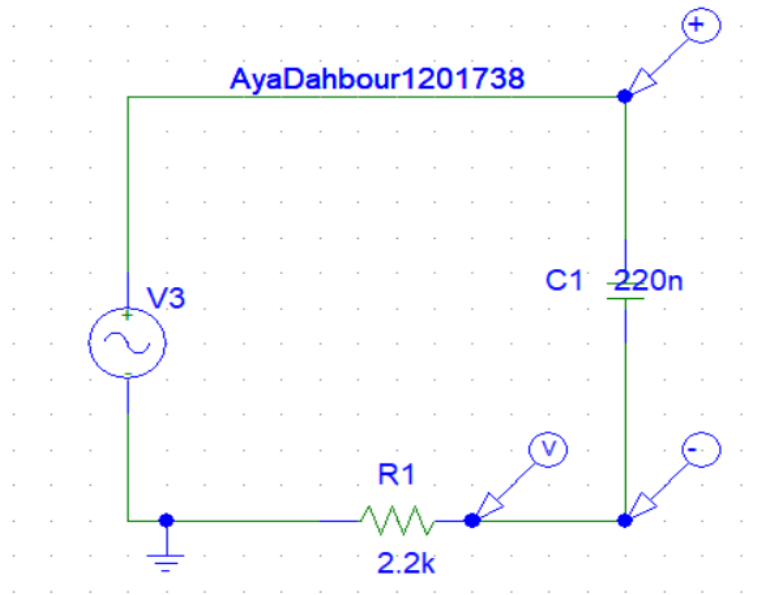
Section: 3

Date: 6/12/2022

A. Passive filter

1.First order circuits:

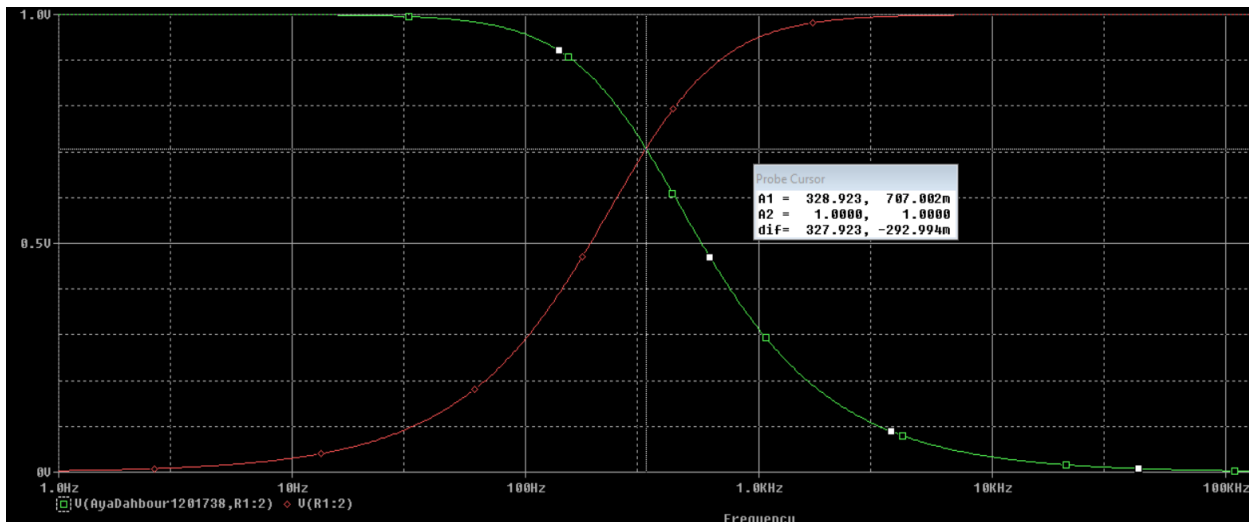
Circuit using PSpice:



CIRCUIT 1

$$F_c = 1/2\pi RC = 1/2\pi(2.2k)(220n) = 328.8\text{Hz}$$

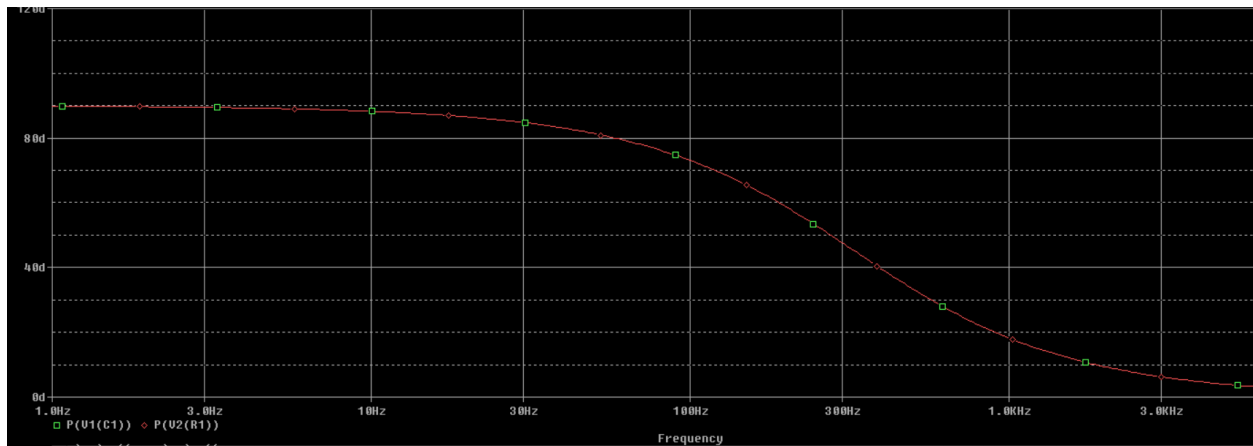
- VR1 and VC1 orders of magnitude in decibels:



first order

According to the above plot, the 3dB cut-off frequency is 327.923 Hz.

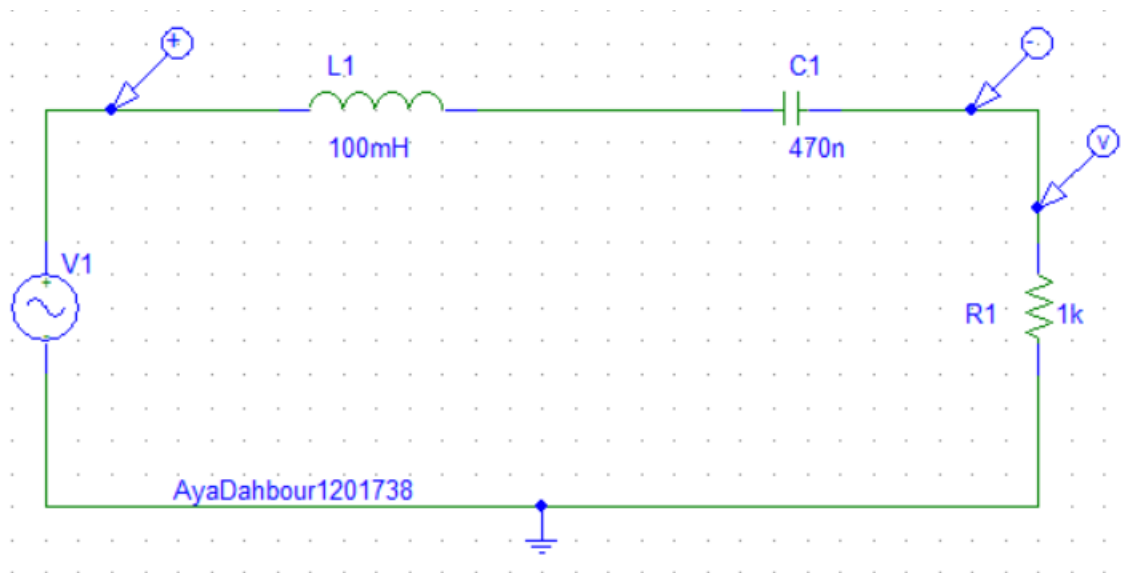
- Plot phase of V_R and V_C in degrees: i.e. : $p(V_R)$ and $p(V_C)$.



$p(V_R)$ & $p(V_C)$.

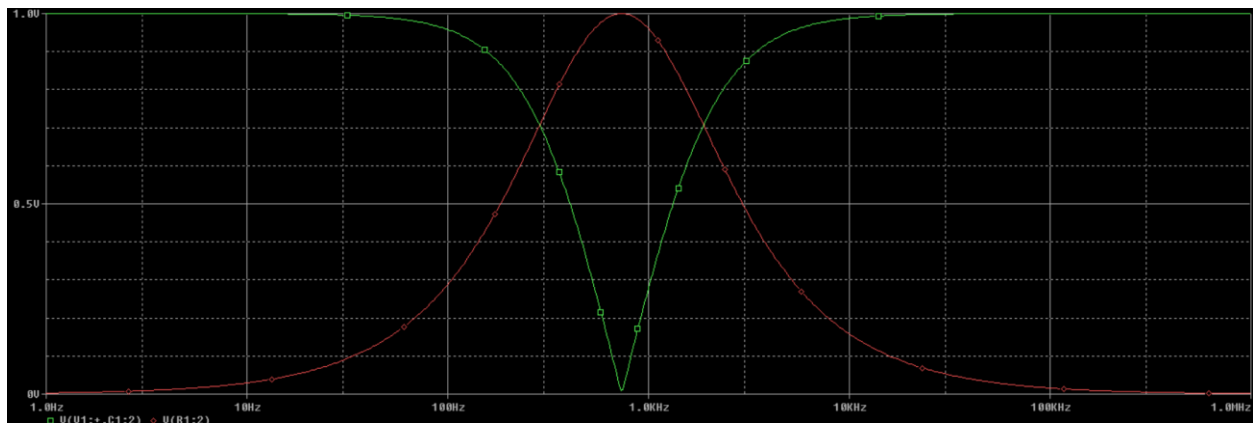
2. Second order circuits:

Circuit using PSpice:



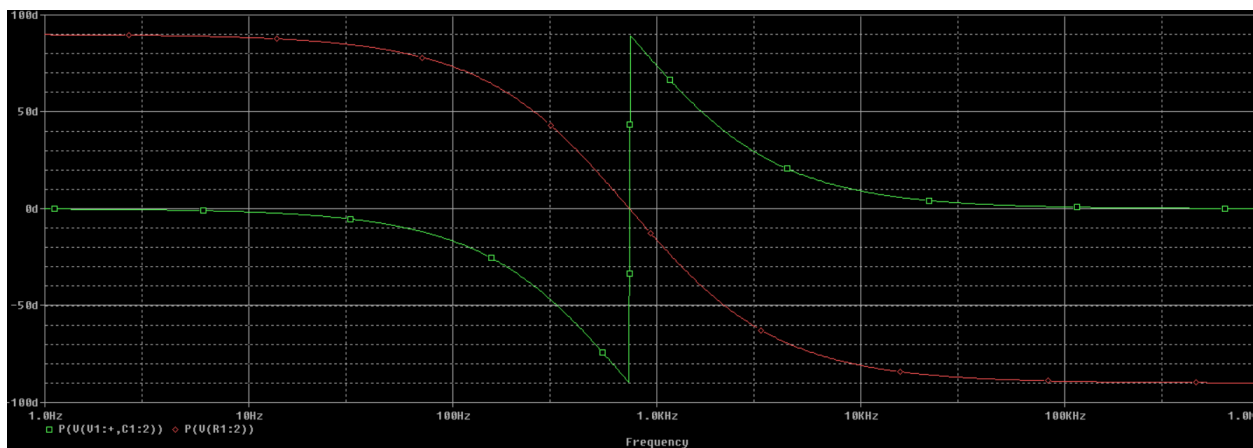
circuit 2

- magnitude of V_R and $(V_C + V_L)$ in decibels:



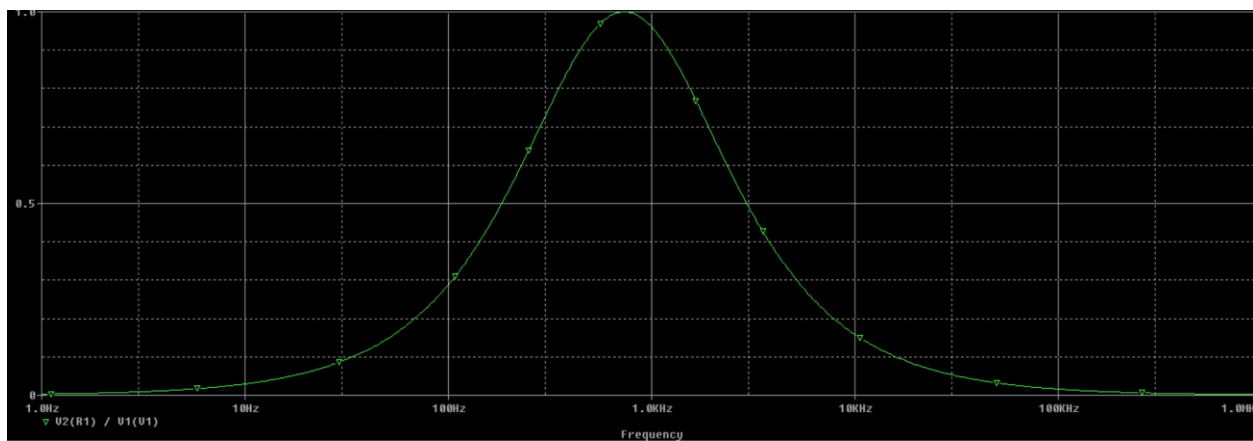
magnitude V_R & (V_c+V_L)

- phase of V_R and (V_c+V_L)



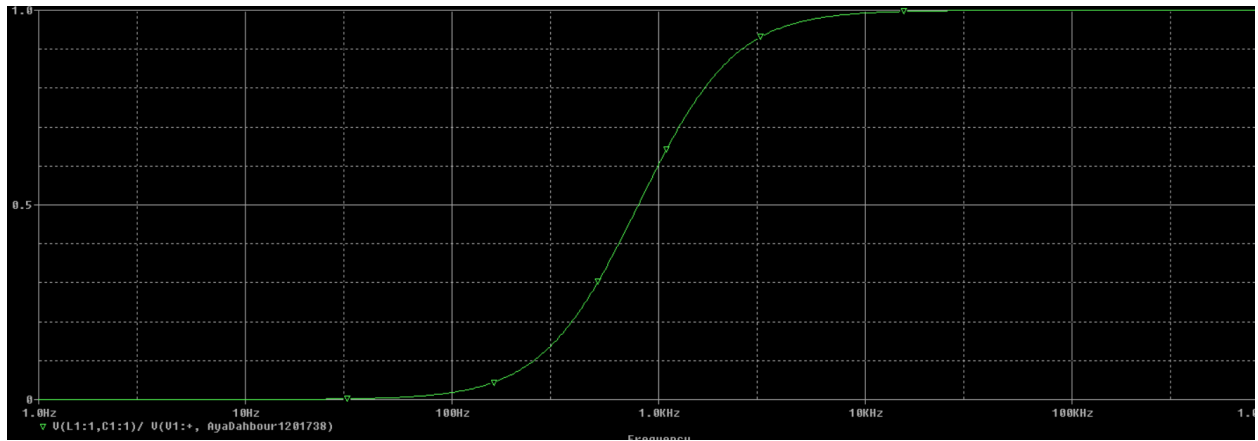
phase V_R & (V_c+V_L)

- From the magnitude-frequency plots of (V_R / V_i) :
➤ Band-pass filter



(V_R / V_i)

- the magnitude-frequency plots of $((V_C + V_L) / V_i)$:

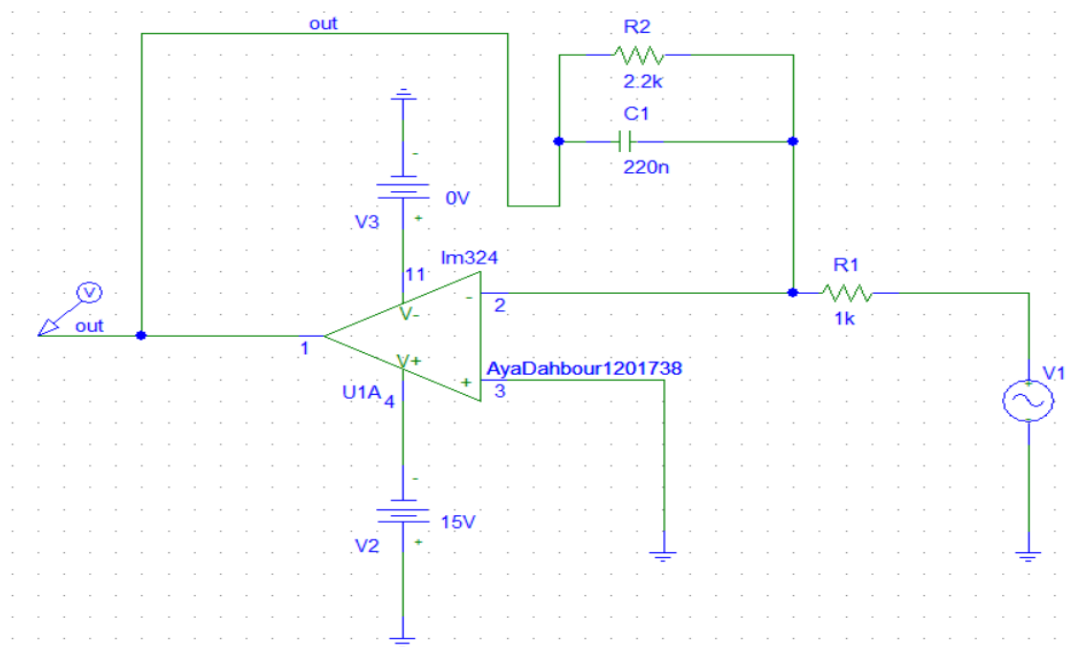


$$((V_C + V_L) / V_i)$$

➤ High-pass filter

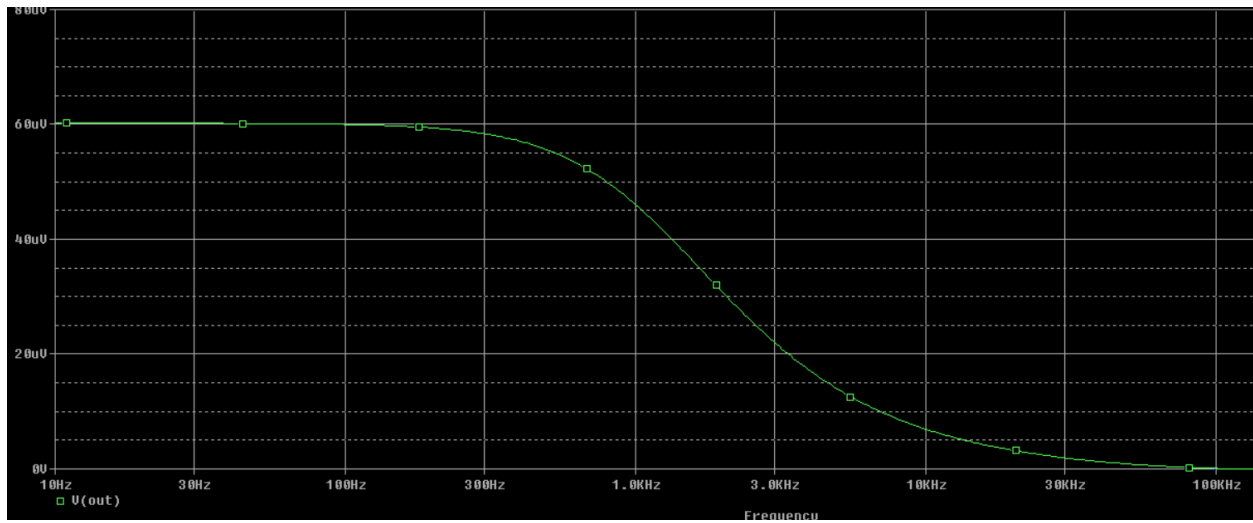
B. Active filters:

Circuit using PSpice:



circuit3

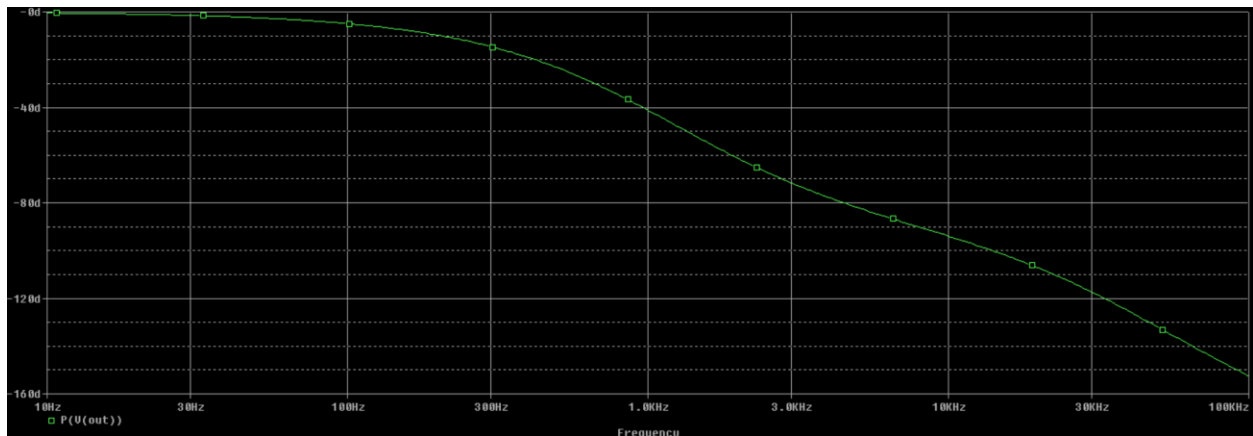
- magnitude of V_o in decibels :



magnitude of V_o

- Low-pass filter
- From theory: $f_c = 1/2\pi R_1 C = 1/2\pi(2.2k)(220n) = 328.833 \text{ Hz}$
- $V_o = R_1/R_2 = 2.2K/1K = 2.2v = 20 \log 2.2 = 6.848 \text{ db}$
- From plot: $V_o = 6.843\text{db}$
- $6.843 - 3 = 3.843\text{db}$ so $f_c = 326.459\text{Hz}$

- phase of V_o :



phase of V_o